

Rajdeep Gill

(431) 998-9785 | hi.rajdeepgill@gmail.com | github.com/Rajdeep-Gill

EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

May 2026

University of Manitoba

Winnipeg, MB

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Parallel Computing, Computer Architecture, Operating Systems, Differential Equations, Partial Differential Equations, Stochastic Processes

Awards: NSERC Undergraduate Student Research Award (2023, 2024), Computer Engineering Academic Excellence Award (2024)

EXPERIENCE

Robinhood

May 2025 - Sept 2025

Software Engineer Intern

Toronto, ON

- Incoming intern for the Summer 2025 term on the Brokerage team

Biosensors Research Lab - University of Manitoba

May 2024 - Dec 2024

Undergraduate Research Student

Winnipeg, MB

- Designed and trained a convolutional neural network using Python and PyTorch to predict cell viability from 10,000+ diffraction images, achieving 85% accuracy, demonstrating potential for non-invasive cell viability assessment
- Created a centroid detection algorithm, improving particle identification accuracy from 75% to 92%
- Developed an interactive data visualization tool with Tkinter to process filter, and display model predictions
- Engineered a SQL database architecture for experimental data management and created a RESTful API with Flask to facilitate data retrieval and analysis

Biosensors Research Lab - University of Manitoba

May 2023 - Aug 2023

Undergraduate Research Student

Winnipeg, MB

- Applied dimensionality reduction using Python and Scikit-learn to analyze and cluster cells based on scattering patterns
- Optimized data analysis pipeline through parallel processing and refactoring existing algorithms, achieving 2.5x speedup
- Developed an automated oscilloscope testing platform with Python and PyVISA, streamlining data collection protocols to enhance measurement speed and ensure repeatability

PROJECTS

Collaborative Document Editor | React, TypeScript, HTML/CSS, Next.js, Socket.io, SQL

- Developed a collaborative document editor supporting real-time collaboration and user authentication
- Utilized Socket.io for real-time collaboration and implemented autocomplete and in-line suggestions using Google Gemini

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js, SQL

- Created an expense and income tracking dashboard with user authentication and easy data visualization
- Implemented a RESTful API to handle user data and transactions, and a PostgreSQL database to store user information

Machine Learning Classification of Microplastics | Python, PyTorch, OpenCV

- Developed a machine learning model to classify microplastics based on their optical and dielectrophoresis features
- Created an image processing pipeline to extract features, discard low-quality images, and prepare data for training
- Paper accepted at IEEE AP-S/URSI 2025, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features"

Distributed Image Processing Pipeline | C++, OpenMPI

- Implemented a parallel image processing framework using OpenMPI that accelerated large image analysis by 3.8x across a 4-node cluster
- Implemented a custom communication protocol to ensure low latency and high throughput

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript

Frameworks & Libraries: PyTorch, OpenCV, OpenMPI, Tkinter, React, Node, Express, Spring Boot, Flask

Tools: Git, Jenkins, Jira, AWS, Linux, Docker, Firebase, SQL, NoSQL, Excel, REST API

Concepts: Backend Development, Frontend Development, Fullstack Development, Software Engineering, Machine Learning, Networking, CI/CD, Microservices, Infrastructure Engineering, DevOps, Automation, Agile